

COSC 0160: DATA COMMUNICATION & COMPUTER NETWORKS (30/30 CF 3.0)

Purpose: To introduce students to fundamentals of data communication and computer Networks

Learning outcomes

At the end of the course, students should be able to;

- Demonstrate knowledge of the working and maintenance of telecommunication networks
- Explain different transmission modes
- Explain network architectures and topologies
- Manage telecommunications
- Identify and manage various security threats

Description/Content

Introduction to computer network LANs, Transmission media, Network hardware, LAN topologies and protocols. Lan design, administration and troubleshooting. Communication concepts. Bit and Baud rates. Synchronous, parallel and serial transmission modes. Frequency and time domains, communication media. Modulation and Demodulation, Signal encoding, sampling theorem, nyquist theory Communication protocols and architectures; error detection and correction .OSI model, T.C.P/IP model, Practical in network installation and configuration

Delivery of the Unit:Lectures: 2 Hours per week: Practical 2 hours per week

Evaluation: Ordinary Examination at end of Semester 70% ;Continuous Assessment 30%

Total: 100%

References:

1. Shay, W. A. (1998). Understanding data communications and networks. International Thomson Publishing.
2. Halsall, F. (1988). Data Communications, Computer Networks.
3. Stallings, W. (2007). Data and computer communications. Pearson/Prentice Hall.
4. Schwartz, M. (1977). Computer-communication network design and analysis (Vol. 25). Englewood Cliffs, NJ: Prentice-Hall.
5. Barber, D. L., & Price, W. L. (1979). Computer networks and their protocols (Vol. 10, No. 11). Chichester: Wiley.

6. Taub, H., & Schilling, D. L. (1986). Principles of communication systems. McGraw-Hill Higher Education.